

JONATHAN KIM

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Education

The University of Texas at El Paso (UTEP)

Expected: December 2026

Bachelor of Science in Computer Science

GPA: 3.73

Relevant Coursework: Operating Systems, Data Structures & Algorithms, Computer Organization, Software Engineering

Experience

PM Technologies LLC

November 2025 – Present

Systems Engineering Intern

El Paso, Texas

- Assembled drone prototypes using 3D-printed frames, motor systems, and power components to accelerate iteration.
- Integrated sensor and compute modules for autonomous flight control, ensuring accurate telemetry for navigation and stability algorithms.
- Supported hardware design for mission-critical drone platforms by debugging integration issues and improving test repeatability.

The University of Texas at El Paso

January 2025 – May 2025

Undergraduate Teaching Assistant - Elementary Data Structures & Algorithms

El Paso, Texas

- Mentored **30+** students in **Java** on data structures, improving lab completion and debugging accuracy.
- Designed debugging guides and walkthroughs on recursion and pointer-based structures.
- Resolved **20+** coding issues weekly by teaching error tracing and logic refactoring.

The Ohio State University AI-EDGE Institute

May 2024 – July 2024

Generative AI Researcher

Columbus, Ohio

- Optimized **Stable Diffusion 2** with **LoRA** training in **PyTorch**, processing **5** datasets to reduce training time by **18%**.
- Fine-tuned **GPT-2** on a **1M-token dataset** and evaluated outputs with **perplexity** and **BLEU**, improving accuracy by **4%**.
- Built data preprocessing pipelines in **Python** to clean, tokenize, and batch training inputs for multiple model experiments.

Personal Projects

AeroScout | Python, FastAPI, OpenCV, YOLOv8, InsightFace, React, WebRTC

October 2025 – Present

- Designed an AI threat-assessment system using **InsightFace**, **YOLOv8**, and **React** to detect weapons and identify persons of interest with **90%** test accuracy.
- Implemented **REST** endpoints in **FastAPI** and **OpenCV** fusion pipelines to combine detections and compute **<200ms** risk scores per frame.
- Developed a **WebRTC React** dashboard for real-time video streaming and alert displays with optimized state handling for consistent low-latency performance.

Hada | TypeScript, React Native, Expo, Firebase

May 2025 – Present

- Built a mobile fitness app in **React Native** that logs workouts, tracks daily streaks, records progress gains, and monitors food intake with real-time state updates.
- Engineered **Firebase Authentication** and **Firestore** data models with structured collections to sync user metrics reliably across devices.
- Evaluated performance with **20+** beta users, resolving sync issues for reliable workout, streak, and nutrition updates.

Fight Coach (Hackathon) | Python, MediaPipe, OpenCV, Hugging Face Transformers, Flask

April 2025

- Developed a real-time pose analysis tool using **MediaPipe** to extract 2D skeletal keypoints from fight footage.
- Refined a transformer-based classifier with **Hugging Face** to detect stances and form flaws across **3K+** pose sequences.
- Engineered a pipeline to preprocess frames, convert landmarks to vectors, and stream time-series data into the model.

DUI Risk Radar (Hackathon 2nd Place) | JavaScript, React, Firebase, HTML/CSS

September 2024

- Led a team of **3** to build a **React + Firebase** web app that mapped DUI incidents and highlighted high-risk driving zones with responsive, real-time updates.
- Implemented **Google Maps API** heatmaps to render **100+** geo-points and visualize density patterns.
- Constructed a **Google Cloud Speech-to-Text** tool for live transcription and structured **Firebase** logging with minimal latency overhead.

Technical Skills

Programming Languages: Python, Java, JavaScript, TypeScript, Swift, SQL, C

Frameworks/Tools: React, Git, React Native, Firebase, AWS, Docker, SwiftUI, Flask, Node.js, REST APIs, PyTorch, Pandas